

HDOC Day-36

String Operations Using a Menu-Driven C Program

Aim

To write a menu-driven C program using separate functions to perform different operations on a string without using built-in string length functions.

- Count the number of characters in a string.
- Print each character of the string on a new line.
- Count the number of vowels and consonants in the string.
- Exit the program.

Algorithm

1. Start the program.
2. Declare a character array to store the string.
3. Accept a string from the user.
4. Display the menu containing the four operations.
5. Accept the user's choice and use a switch statement to perform the selected operation.
6. For Count Characters, traverse the string until '\0' is encountered and count every character.
7. For Print Each Character, traverse the string and print every character on a separate line.
8. For Count Vowels and Consonants, examine alphabetic characters and count a, e, i, o and u (uppercase or lowercase) as vowels; count other letters as consonants. Ignore spaces, numbers and special characters.
9. For Exit, terminate the program.
10. Display an appropriate message for an invalid menu choice.
11. Continue displaying the menu until Exit is selected.
12. Stop the program.

C Program

```
#include <stdio.h>

void countCharacters(char str[]);
void printCharacters(char str[]);
void countVowelsConsonants(char str[]);

int main()
{
    char str[200];
    int choice, i = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    /* Remove newline added by fgets */
    while (str[i] != '\0')
    {
        if (str[i] == '\n')
        {
            break;
        }
        i++;
    }

    while (1)
    {
        printf("\nMenu:\n");
        printf("1. Count Characters\n");
        printf("2. Print Each Character\n");
        printf("3. Count Vowels and Consonants\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        int choice;
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                countCharacters(str);
                break;
            case 2:
                printCharacters(str);
                break;
            case 3:
                countVowelsConsonants(str);
                break;
            case 4:
                return 0;
            default:
                printf("Invalid choice. Please try again.\n");
        }
    }
}
```

```

        str[i] = '\0';
        break;
    }
    i++;
}

do
{
    printf("\n----- STRING OPERATIONS MENU ----- \n");
    printf("1. Count Characters\n");
    printf("2. Print Each Character on a New Line\n");
    printf("3. Count Vowels and Consonants\n");
    printf("4. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            countCharacters(str);
            break;

        case 2:
            printCharacters(str);
            break;

        case 3:
            countVowelsConsonants(str);
            break;

        case 4:
            printf("Exiting program...\n");
            break;

        default:
            printf("Invalid choice! Please enter a number from 1 to 4.\n");
    }

} while (choice != 4);

return 0;
}

void countCharacters(char str[])
{
    int i = 0, count = 0;

    while (str[i] != '\0')
    {
        count++;
        i++;
    }

    printf("Number of characters = %d\n", count);
}

void printCharacters(char str[])
{
    int i = 0;

    printf("Characters in the string:\n");

    while (str[i] != '\0')
    {
        printf("%c\n", str[i]);
        i++;
    }
}

void countVowelsConsonants(char str[])
{
    int i = 0;
    int vowels = 0, consonants = 0;

    while (str[i] != '\0')

```

```

{
    if ((str[i] >= 'A' && str[i] <= 'Z') ||
        (str[i] >= 'a' && str[i] <= 'z'))
    {
        if (str[i] == 'A' || str[i] == 'E' ||
            str[i] == 'I' || str[i] == 'O' ||
            str[i] == 'U' || str[i] == 'a' ||
            str[i] == 'e' || str[i] == 'i' ||
            str[i] == 'o' || str[i] == 'u')
            vowels++;
        else
            consonants++;
    }
    i++;
}

printf("Number of vowels = %d\n", vowels);
printf("Number of consonants = %d\n", consonants);
}

```

Test Case Table

Test Case	Operation	Input	Expected Result
1	Count Characters	Hello	Characters = 5
2	Count Characters	Hello World	Characters = 11
3	Print Characters	CAT	C, A, T on separate lines
4	Vowels & Consonants	Education	Vowels = 5, Consonants = 4
5	Vowels & Consonants	Hello World	Vowels = 3, Consonants = 7
6	Vowels & Consonants	C Programming	Vowels = 3, Consonants = 9
7	Invalid Choice	Choice = 5	Invalid choice message
8	Exit	Choice = 4	Program exits

Sample Output

Enter a string: Hello World

```

----- STRING OPERATIONS MENU -----
1. Count Characters
2. Print Each Character on a New Line
3. Count Vowels and Consonants
4. Exit

```

Enter your choice: 1
Number of characters = 11

Enter your choice: 3
Number of vowels = 3
Number of consonants = 7

Enter your choice: 4
Exiting program...

Result

The menu-driven C program was successfully implemented using separate functions, strings, loops, conditional statements, and a switch-case statement. It counts characters without using built-in length functions, prints each character on a separate line, counts vowels and consonants, and continues until the Exit option is selected.